KAPASA MAKASA UNIVERSITY

ENTOMOLOGY

UNIT EIGHT

TOPIC: INSECT ECOLOGY AND BEHAVIOUR

Objectives: by the end of this unit, SSBAT;

- Population Dynamics
- Describe insect feeding
- Explain survival strategies
- Explain insect defence mechanism

Survival strategies

Insects have developed elaborate behaviour and adaptation strategies that have contributed to their survival. They have very high reproduction rates, (great many offspring). This increase the chance that the offspring will survive to adulthood. Climate and weather changes not only affect the status of insect pests but also affect their population dynamics, distribution, abundance, intensity and feeding behaviour.

Insect survival strategies should be said to have been implemented or practiced when insects have to;

a. Manage to hide from Rainstorms, Windstorms, Ice storms and hail storms
b. Find water in the desiccating heat of the mid-summer
c. Keep from freezing in the winter
d. Avoid floods, wildfires and mud slides
e. Locate a suitable food supply (host plant or animal)
f. Elude birds, spiders, mantids, frogs and other predators
g. And escape being impaled on a pin in a student's bug collection.

The following are the most common survival strategies for insects:

1. **Migration**
   At some time in their life cycle many, perhaps most, of these insects engage in a period of directional movement that carries them beyond the range of their local habitat. The long distant insect movement is called Migration. This is a survival strategy with at least six potential advantages:

   a. Escape from natural enemies
   b. Find more favourable growing conditions
   c. Reduce competition or relieve overcrowding
   d. Locate new (or unoccupied) habitats
   e. Disperse to alternate host plants
   f. Reassert the gene pool or minimize in breeding.

Although most insects migrate by flying a few species of insects, such as chinch bugs, Mormon crickets and army worms, to mention but a few, travel on the ground. Migration by flight is often aided by prevailing winds. Larger insects, like dragon flies and monarch butterflies may control the direction of their migratory flights, but smaller insects are carried wherever the wind blows. Migration can be a very risky venture. In most species more than 90% of a population may die in transit. Despite such high mortality rates, the reproductive success of individuals who survive the trek apparently makes the gamble worthwhile for the species as a whole.

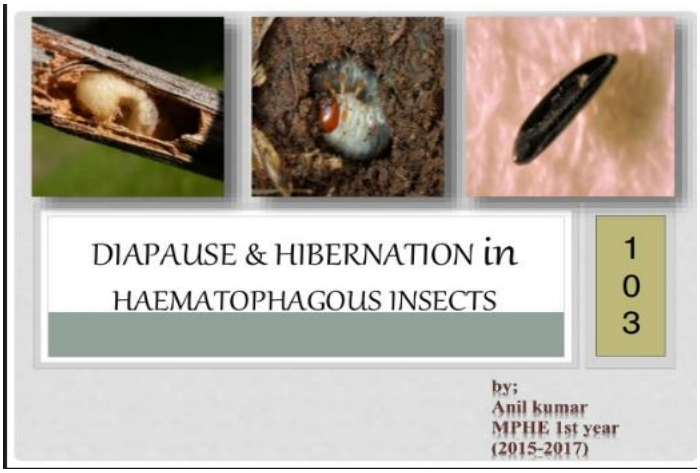


Monarch butterflies roosting on migration in Texas

Diapause

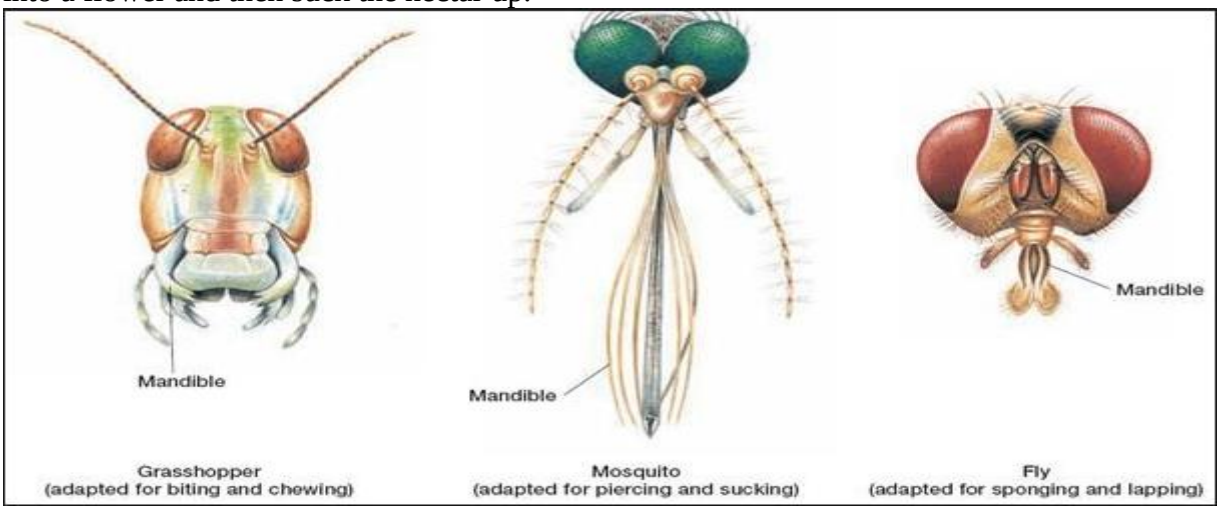
Slow-down of insects' metabolism: a period during which the metabolism of some insects and/or animals slow down, temporarily suspending their bodily development and growth. Such periods are linked to seasonal or environmental changes. Diapause in insects is characterised by

a reduction in oxygen consumption, metabolic rate and physical activities. Feeding and growth are generally interrupted while the individual survives on stored food reserves. Diapause typically occurs during the egg stage in some species, during a nymph or larva/instar in other species or during the pupal stage in still other species. Even adults may enter a reproductive diapause which causes a significant delay in the onset of sexual maturity.



Specialized Mouth Parts

Many insects have specialized ways of eating food that helps them to survive. Some have special mouthparts for chewing, such as grasshoppers. These mouthparts have mandibles, or jaws, that move sideways to allow the insect to bite and chew their food. Other insects have mouthparts that are ideal for piercing or sucking food, such as the cicada. These insects pierce their food with needle-like stylets and then suck the juice into their mouths. Some insects have sponging mouthparts, such as the housefly. These insects lap up food by way of two labella, or soft lobes. The liquid then travels through a thin tube so the animal can eat it. Another type of mouthpart used by certain insects is the chewing/lapping mouthpart. Insects such as the honeybee have such mouthparts that can be used both for biting and sucking food. Honeybees, for example, first bite into a flower and then suck the nectar up.



Camouflage and Mimicry

The ability to hide from predators who would like to eat them and the ability to hide within their habitats so that they will be able to eat, reproduce, and protect their young have contributed to insect's success in surviving. Some insects use camouflage in order to escape the notice of predators. Camouflage is a way to hide. Some insects have bodies that are the same colour as their surroundings, for example, so they blend in with their environment. Other insects mimic, or imitate, other animals. Mimicry is an adaptation in which an animal is protected from a predator due to its likeness to another unpleasant animal. Insects may be brightly coloured so that other animals can see them and know they are dangerous or poisonous and will avoid them. The Ash Borer moth is brightly coloured like a wasp, although it is not capable of stinging as the wasp is. Since its colours look like a wasp, however, animals that have learned to avoid wasps due to their colours will likewise stay clear of the Ash Borer moth. When the mimic does not share the dangerous trait of the animal it is mimicking, it is called Batesian mimicry, named for the scientist Henry Walter Bates, who first noticed this form of mimicry. Some insects use a type of mimicry called auto-mimicry, where one part of the body on the insect resembles another part of the body. Some caterpillars, for instance, have markings on their tails that look just like the markings on their eyes. This is useful in case a predator is attacking them, because the predator may attempt to eat the tail of the caterpillar instead of its head, which would be far less dangerous to the insect. A few examples of other unique arthropod adaptations are described below:

- Butterflies can be difficult to catch because of their loose-fitting wing scales which make them slippery to a predator
- Wasps can paralyze caterpillars with their sting
- The American spider has a ladder-like web that knocks the scales off a moth so that it sticks to the silk
- Caterpillars feed on poisonous plants which serve as a chemical protection against larger animals such as birds
- Butterflies and moths can copy the warning patterns and colours of the foul tasting, toxic varieties (mimicry)
- To avoid being spotted by predators, many species blend into bark, leaves, flowers, and twigs (camouflage)
- Some species of caterpillars can resemble bird droppings to keep them safe (camouflage)
- Sphinx moths have eye-like spots behind the head or on the underside of the body; if suddenly touched, the caterpillar hides its real head and brings these spots into prominence so that it looks like a snake (mimicry)
- Birds will usually try to take their first lethal peck at their prey's head, having been attracted to the eyes; some butterflies have false heads on their hindwings (auto-mimicry)



3. Mimicry

Another example of is the locust borer. This insect not only looks like a bee or wasp, it sounds like one, too!



